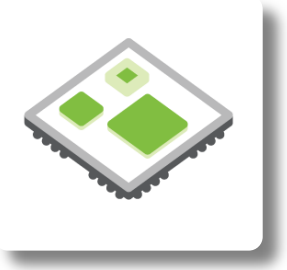
SERVER:
OPEN CHIPLET
ECONOMY

Securing the Open Chiplet Economy: Defining a Common Security Architecture for FCSA





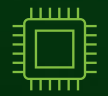
Dominic Brown *Senior Principal Engineer, Arm*

Sylvain Guilley *CTO & Co-founder, Secure-IC*

The Road to an Open Chiplet Ecosystem

Design Reuse

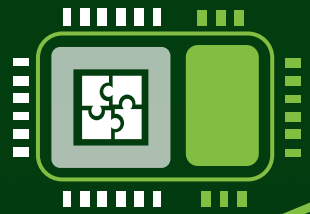
Monolithic System



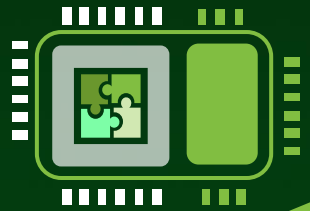
Chiplet-based SoC's with Production-ready IP

Single-generation Design Reuse

Proprietary Multi-die System



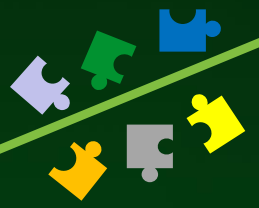
Semi-Custom



Chiplet-based SoC's with Production-ready IP

Multi-generation Design Reuse

Merchant Chiplets



Single Die, Multiple Systems, Design Reuse

- Accelerated design timelines
- Reduced investment
- Broader SKU options

Foundation Chiplet System Architecture Vision

Define Chiplet Types

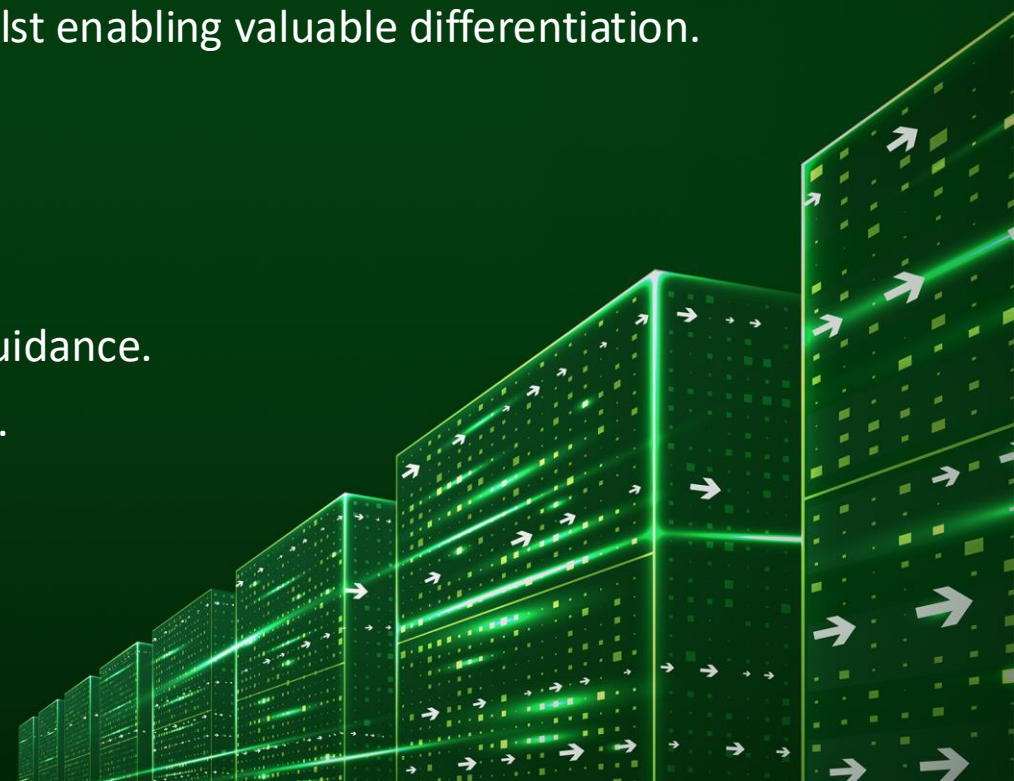
- Classify chiplet types, properties, and functional requirements within a system.
- Establish a common language to enable the industry to collaborate on multi-die systems.
- Identify system-level chiplet interfacing requirements.

Interoperability

- Create consistent chiplet interfaces and connectivity, whilst enabling valuable differentiation.
- Simplify chiplet specification and integration.
- Establish mutual expectations between chiplets.

Reuse

- Establish chiplet interface compatibility and versioning guidance.
- Enable chiplets to be designed independently of systems.
- Accessible to both system and chiplet designers.



Foundation Chiplet System Architecture

- FCSA v1.1 is available now.
- It contains:
  - A new security chapter.
  - Additional interface definition and rules for boot sequencing.
  - Additional rules on implementation requirements.
  - Various minor fixes and improvements.

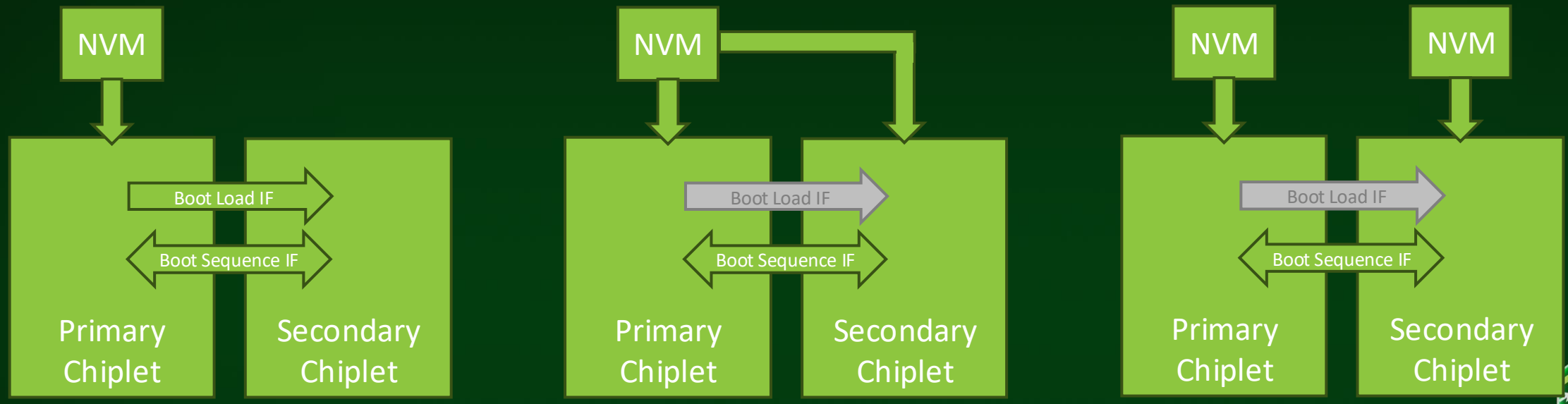


Boot Sequencing

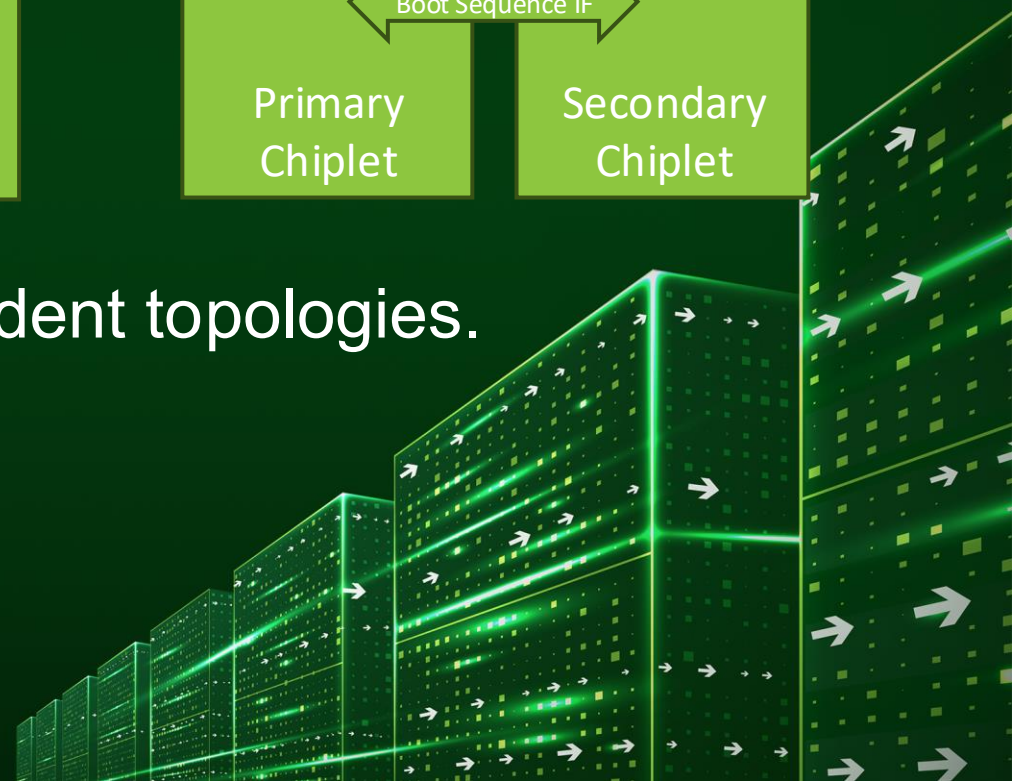
- The changes for v1.1 include:
  - The specification of an interface for boot sequencing.
  - Detail of boot sequencing between trusted security agents and system controllers.
  - Rules as appropriate for chiplet types



Boot Sequence Control Interface



- Supports Distributed, Shared, and Independent topologies.
- Common signal set regardless of topology.



Implementation Requirements

- A small new chapter that captures interoperability and scalability requirements.
  - Adapting memory traffic across differing chiplet interfaces.
  - Size snoop filters for total system demand.
- This chapter creates a place to add more requirements of this kind.



Security Policies & Governance

- Chiplet-based systems are:
  - multi-vendor, multi-configuration, field-updatable, and deployed in diverse markets.
    - For example, automotive, IoT, datacenter, defense.
- Security is context dependent:
  - Different acceptable risk levels, trust models, regulations, priorities (availability, safety, integrity)

→ Govern trust decisions and system behavior based on context-dependent policies.

- Decisions are based on inputs and policy rules:
  - System RoT collects inputs from chiplets CRoT-A and applies policy.

Goals of Security in FCSA v1.1

- **GOAL 1: Completeness of the analysis.**
- **METHOD**:
  - Root the threat analysis and risk assessment into a general scope with terms of reference, that is supportive of reasoned deduction from the TARA.

- **GOAL 2: Position as a milestone for achieving compliance**.
- **METHOD**:
  - Make the chapter a template for the derivation of requirements compatible with existing frameworks.
  - Enumerate formal requirements that are testable.

Chapter on the Security of Chiplet Systems

Security Objective

SoC-equivalent security for chiplet systems: a chiplet system must be able to provide the same level of security as an equivalent monolithic SoC implementation.

Chiplet System Security Model

- Definition of concepts and components of a chiplet system that are used for security analysis of a design.
- Definition of adversarial models that the system designer might consider.

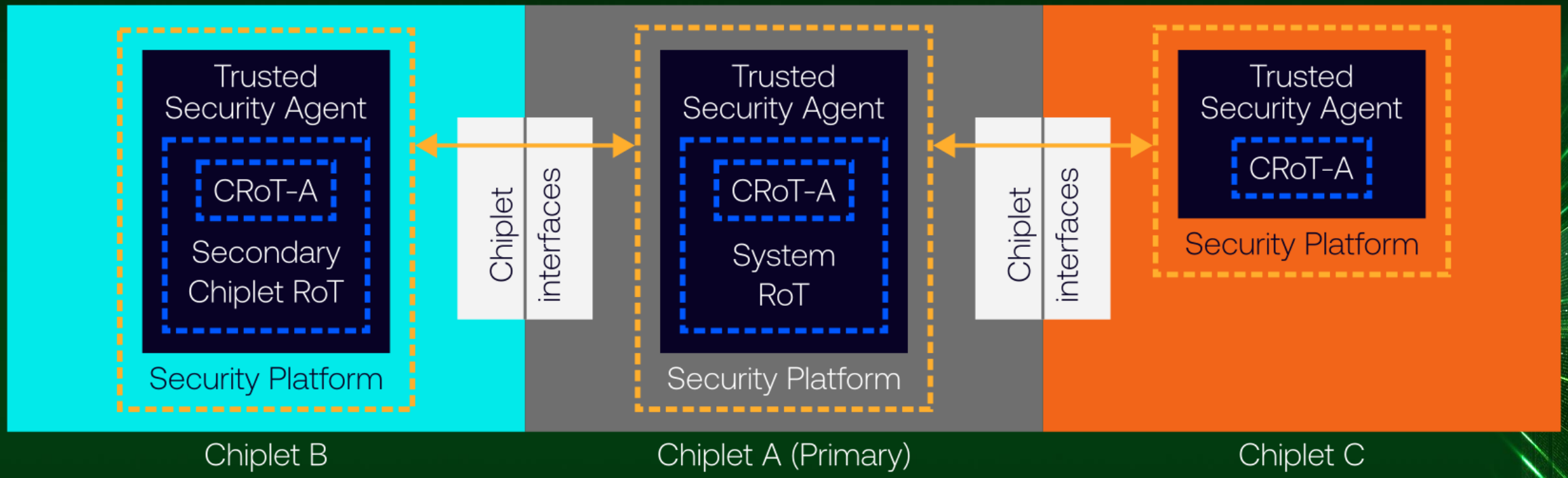
Threats & Mitigations

- A methodology for security risk assessment that the system designer applies.
- An exhaustive list of threats.
- A mapping of mitigations for those threats.

Future work

- Minimum specification for profiles.
- Complete-as-possible specification of implementation for interoperability.

Chiplet System Security Model



- Important context is identified: assumptions about chiplet systems and their supply chain, stakeholders, security lifecycle and security goals.
- Also defined are chiplet system topology models and adversarial models, used later for the risk assessment methodology.

Threat Analysis and Risk Assessment (TARA)

General remarks

- Explicit application **ISO/SAE 21434:2021 TARA** methodology
  - Assets Identification
  - Threat Scenario Identification
  - Impact Rating
  - Attack Path Analysis
  - Attack Feasibility Rating
  - Risk Value Determination
  - Risk Treatment Decision
- Required for **UN ECE R.155** regulatory compliance in the context of automotive



Other relevant threats

- Other relevant threats to take into consideration
  - **DoS/DDoS**
    - Unresponsive Chiplet
  - **TOCTOU**
    - A chiplet or component is verified as trustworthy at one moment but is modified, swapped, or behaves maliciously before or during actual use.
  - **Over-building**
    - Unauthorized production of extra, legitimate-looking Chiplets beyond approved volumes, bypassing traceability and security provisioning and enabling stealth supply-chain compromise.

FCSA and SiP Protection Profile (PP)



FCSA	Security Objectives from PP	
Secure Development	OE.Process-Sec-IC	Maintains secure operation before self-protection becomes active. Operational environment ensures secure development processes.
Vertical Integration	Not supported by the current version of the PP. It can be added as an assumption instead of security objective.	
Chiplet Identity	O.Identification	Provides unique identification and secure storage for initialization data.
Chiplet Attestation	O.Trusted_Chiplet	Ensures only authenticated and verified chiplets are integrated.
Chiplet Life Cycle	OE.Iss-Cert O.Anti-counterfeiting	Supports secure metadata signing and issuance during provisioning. Each ownership transfer event triggers binding rule changes that revoke prior owner's chiplet access and establish new owner's authority. Chiplet activation is exclusive to actual owner through enforced user-subject bindings.
Debug Control	To be added.	
System Manifest	O.Identification	Binds chiplet identity and initialization securely for system traceability. Identity of the HBOM, meaning one identifier per chiplet.
Secure Provisioning	OE.Secure-Initialization	Applies security procedures during chiplet provisioning.
Authorized Update	O.Secure-State	Enforces secure state boot and response to tampering or corrupted boot data.
Firmware Measurement		
Secure Chiplet Boot		
Protected Connection	O.Leakage O.Manipulation	Ensures confidentiality and resistance to physical or side-channel leakage during transit or at rest. Detects/responds to manipulation of die-to-die communication.
Chiplet Topology		
Chiplet RoT for Attestation (CRoT-A)	O.Trusted_Chiplet	Manages chiplet authenticity, config control, and integrity of initialization and identity data and ensures only authenticated and verified chiplets are integrated.

SFRs/SARs Examples (O.Trusted_Chiplet)

FAU_SAS.1 **Audit storage**

Hierarchical to:	No other components.
Dependencies:	No dependencies.
FAU_SAS.1.1:	The TSF shall provide the test process before TOE Delivery with the capability to store [selection: TOE unique identification data, Initialization Data, Pre-personalization Data, [assignment: other data]] in the [assignment: type of persistent memory].

FIA_USB.1 **User-subject binding**

Hierarchical to:	No other components.
Dependencies:	FIA_ATD.1 User attribute definition.
FIA_USB.1.2	The TSF shall enforce the following rules on the initial association of user security attributes with subjects acting on the behalf of users: [assignment: rules for the initial association of attributes].
FIA_USB.1.3	The TSF shall enforce the following rules governing changes to the user security attributes associated with subjects acting on the behalf of users: [assignment: rules for the changing of attributes].

FPT_INI.1 **TSF initialization**

Hierarchical to:	No other components.
Dependencies:	No dependencies.
FPT_INI.1.1	The TOE shall provide an initialization function which is self-protected for integrity and authenticity.
FPT_INI.1.2	The TOE initialization function shall ensure that certain properties hold on certain elements immediately before establishing the TSF in a secure initial state as specified in the table below:

Properties	Elements
Correct configuration and initialization of	Master and active chiplets
Integrity of	SBoM

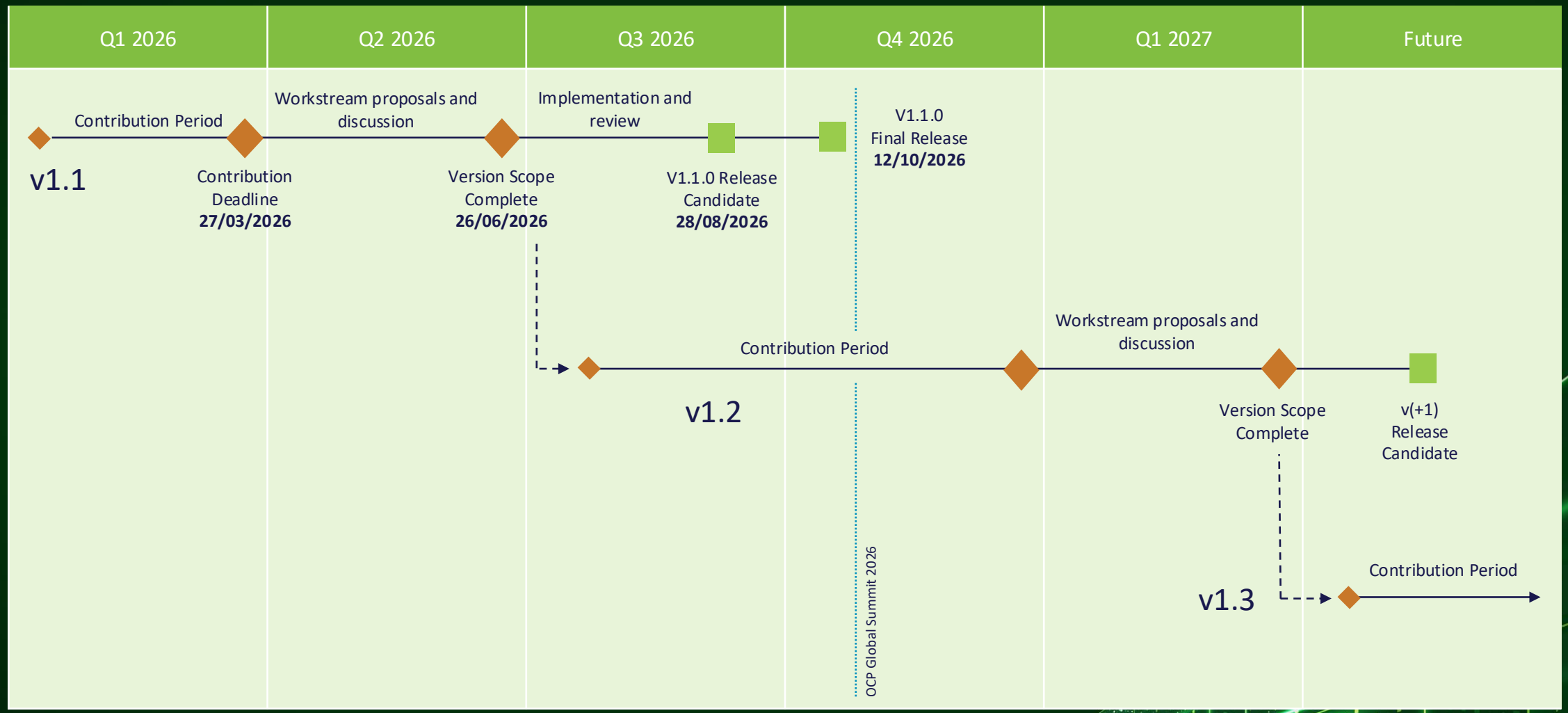
SFRs

ALC: Life-cycle support

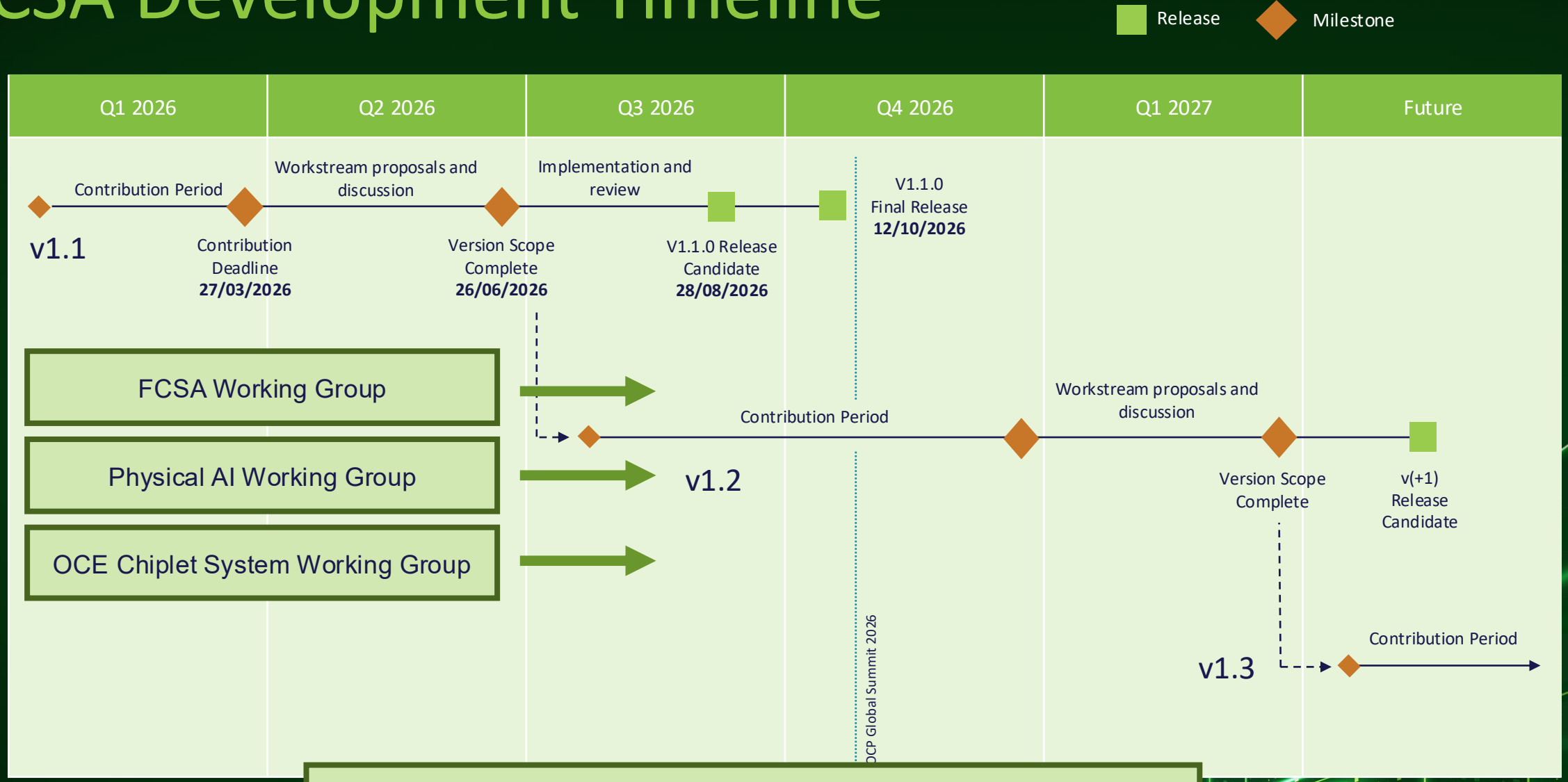
ALC_CMC.4	ALC_CMS.4	ALC_DEL.1	ALC_DVS.1	ALC_FLR.1	ALC_LCD.4	ALC_TAT.1
Production support, acceptance procedures and automation	Problem tracking CM coverage	Delivery procedures	Identification of security measures	Basic flaw remediation	Developer defined life-cycle model	Well-defined development tools

SARs

FCSA Development Timeline



FCSA Development Timeline



Thu, October 15, 2:45pm - 3:05pm | SJCC - Concourse Level - 211

Introduction to a new OCP FCSA Sub-Project Workstream

Topics Under Consideration for v1.2

- Security (More!)
- Functional Safety
- Management Processor Communication
- Time Synchronization
- RAS
- Alignment with the AMBA C2C protocol
- Your contributions…!

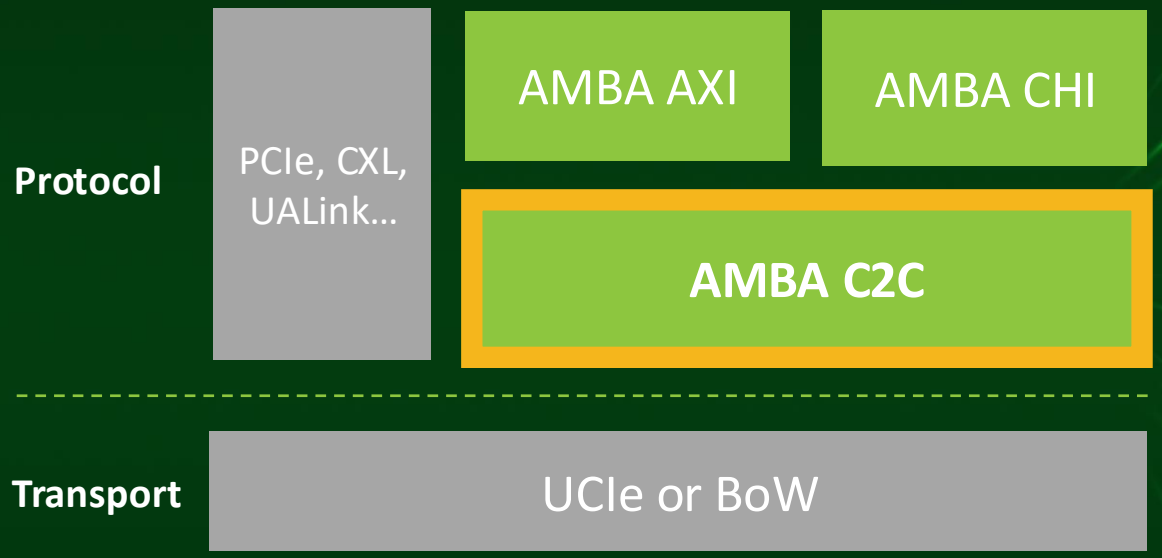


AMBA C2C unifies AXI and CHI across chiplet links

WHY IT MATTERS

1. **Preserve protocol choice**
   AMBA AXI and AMBA CHI can coexist across chiplet architectures
2. **Ensure interoperability**
   AMBA C2C provides a common packetization layer
3. **Optimise the physical link**
   Map bus traffic efficiently onto standard C2C transports and physical layers

A COMMON LAYER ENABLING MAXIMUM COMPATIBILITY



Call to Action

Engage in the FCSA workstream

Contribute technical solutions for review by the community

Review submissions to create a stronger open chiplet ecosystem

https://www.opencompute.org/documents/fcsa-1-1-0-pdf

https://www.opencompute.org/projects/server/open-chiplet-economy/foundation-chiplet-system-architecture/



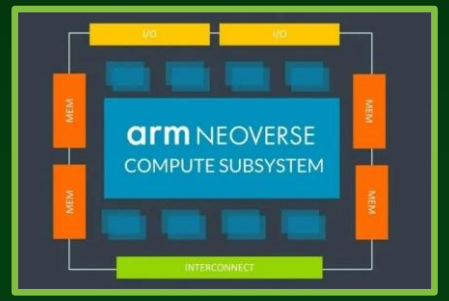

Thank
You!
OCP
GLOBAL
SUMMIT
October 12-15, 2026
San Jose, California

Back Up



Chiplet System Architecture Scope

System design | Functional partitioning | Physical design



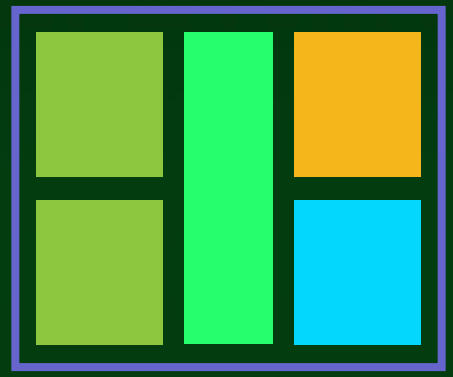
System software

-

Accelerator functionality

-

Compute performance



Chiplet functionality

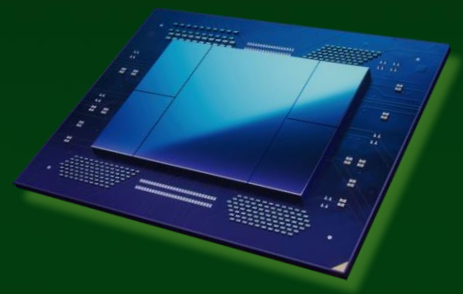
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Protocols

Functional interfaces

-

Physical interface types



Chiplet form factor

-

Physical interface capacity

-

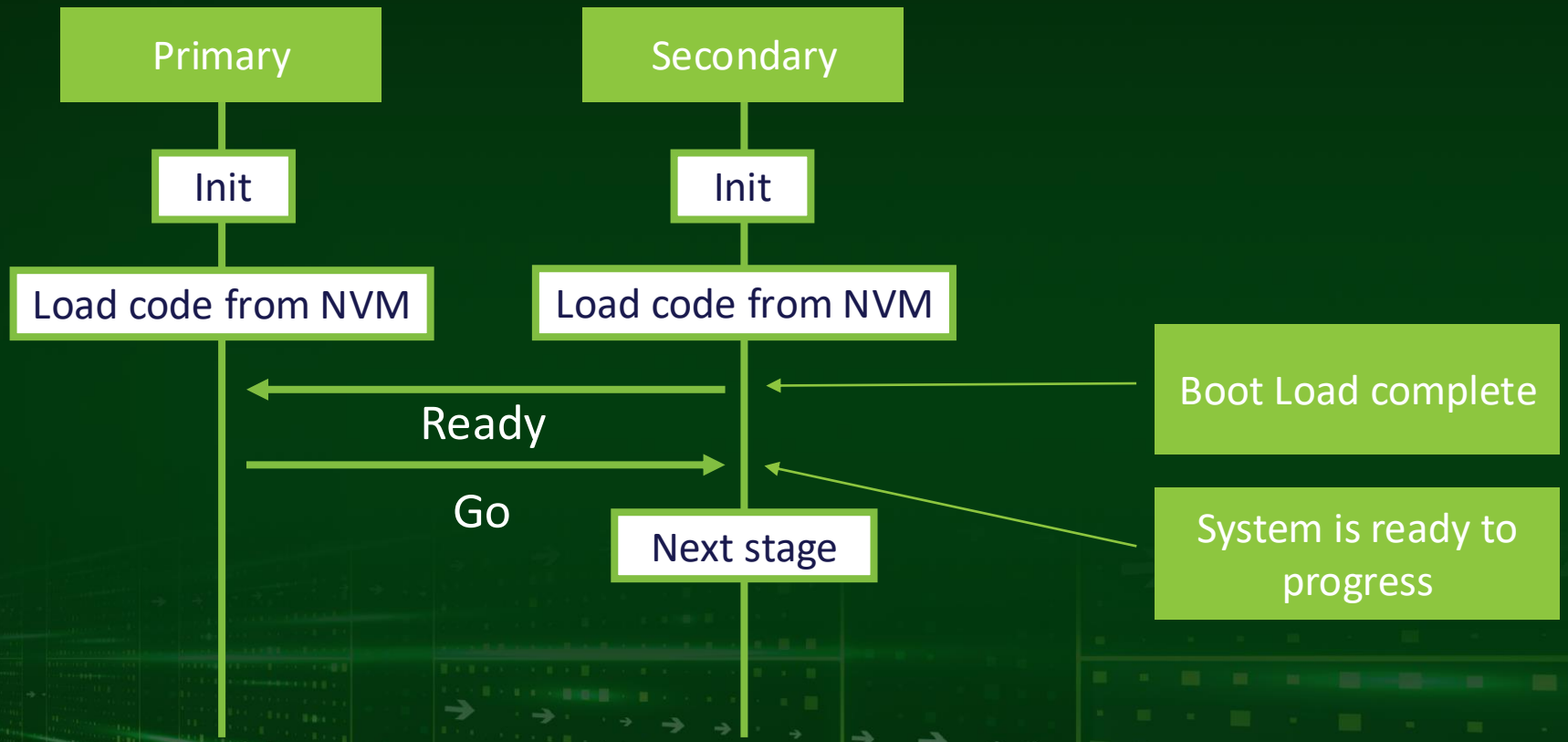
Package design

SCOPE

Scope

Independent Boot Load Sequence Scenario

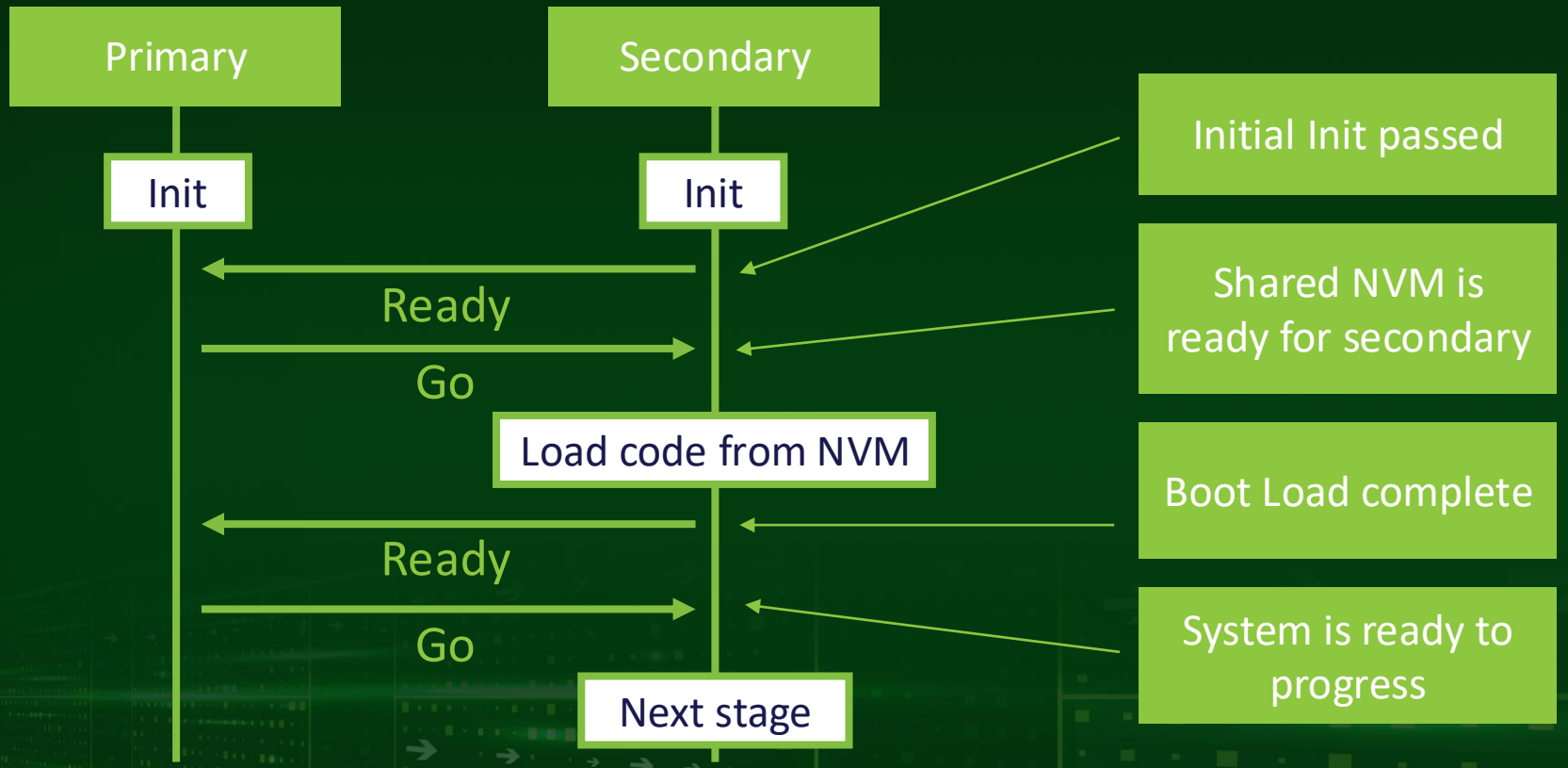
Primary and Secondary have independent NVMs



- Both chiplets load from a separate NVM, then the secondary indicates ready. Once the full system is ready to proceed the primary indicates ”Go”.

Shared Boot Load Sequence Scenario

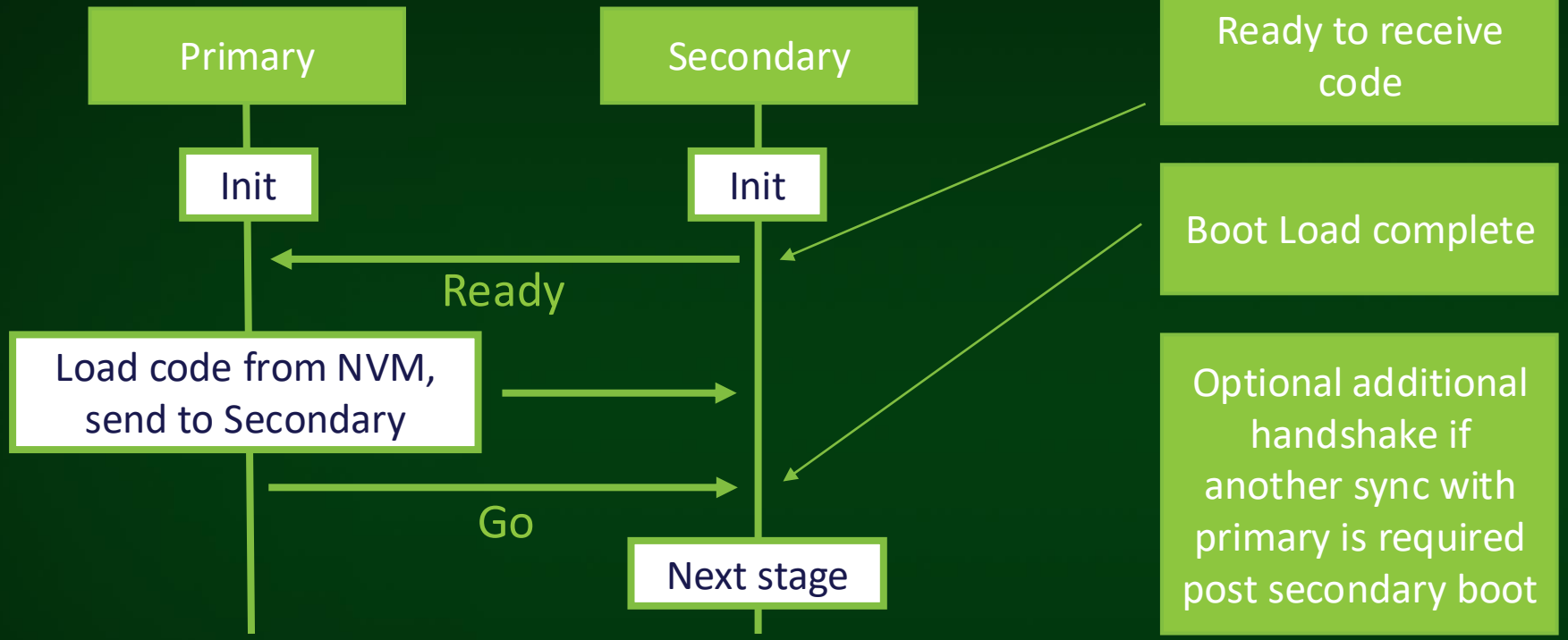
Secondary loads own boot code from shared NVM



- After receiving the “Go” signal, the Secondary entity loads the next boot code image from NVM, then waits for another “go” signal before executing it.

Distributed Boot Load Sequence Scenario

Primary loads boot code to Secondary



Initial Init passed, Ready to receive code

Boot Load complete

Optional additional handshake if another sync with primary is required post secondary boot

- The Primary entity loads a boot code image to the Secondary entity.
- After receiving Go signal, the Secondary executes the boot code.